

Akash S Kumbar

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OVERVIEW

I build learning systems that turn 3D perception into robotic action. My research focuses on spatial intelligence: learning representations of the world that support reconstruction, exploration, and navigation. I am interested in how robots recover meaningful structure from incomplete observations and use that understanding to guide their actions. I take these ideas through the full development cycle, from mathematical formulation and network design to training, evaluation, sensor integration, and deployment on physical robots.

EDUCATION

- IIIT-Hyderabad** Hyderabad, India
MS(Research), Computer Science Engineering; CGPA: 9.33/10.0
2024 – present
- KLE Technological University** Hubballi, India
B.E., Electronics and Communication Engineering; CGPA: 8.42/10.0
2019 – Jul 2023

PUBLICATIONS

* indicates equal contribution.

- Beyond Frontiers: Scene-Anomaly Guided Autonomous Exploration** ↗ : IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
Authors: Akash Kumbar, Abhinav Raundhal, Madhava Krishna.
We introduced SCAGE, which uses learned geometric anomalies in point clouds to select viewpoints for autonomous exploration. Across 10 HM3D evaluation scenes with simulated depth, it achieved **89.7% mean volumetric coverage** and **approximately 70% lower mean Chamfer distance than FrontierNet**; we also demonstrated the system on a physical robot.
- Leveraging 2D Priors and SDF Guidance for Dynamic Urban Scene Rendering** ↗ : Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
Authors: Siddharth Tourani, Jayaram Reddy, Akash Kumbar, Satyajit Tourani, Nishant Goyal, Madhava Krishna, N Dinesh Reddy, Muhammad Haris Khan.
We coupled signed distance functions with 3D Gaussian Splatting and image-derived depth and tracking priors to reconstruct dynamic urban scenes without ground-truth object-motion annotations. The LiDAR-free variant achieved **30.32 dB novel-view PSNR** on NOTR's Waymo dynamic-32 split, exceeding OmniRe's 29.35 dB.
- ASUR3D: Arbitrary Scale Upsampling and Refinement of 3D Point Clouds Using Local Occupancy Fields** ↗ : Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2023.
Authors: Akash Kumbar, Tejas Anvekar, Ramesh Ashok Tabib, Uma Mudenagudi.
We learned local occupancy fields for arbitrary-scale point-cloud upsampling using a single trained model. At 4× upsampling on noisy PU-GAN inputs ($\tau = 0.02$), ASUR3D reduced **point-to-surface error by 8.3% relative to Grad-PU**, with additional validation on captured heritage geometry.
- TP-NoDe: Topology-Aware Progressive Noising and Denoising of Point Clouds Towards Upsampling** ↗ : Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2023.
Authors: Akash Kumbar*, Tejas Anvekar*, Tulasi Amitha Vikrama, Ramesh Ashok Tabib, Uma Mudenagudi.
We combined density-aware neighborhoods, topology-aware perturbations, and a pretrained denoiser to progressively upsample point clouds **without task-specific training**. On the PU-GAN benchmark, TP-NoDe reduced **Chamfer distance by 54% relative to PU-Net**.

EXPERIENCE

- Robotics Research Center, IIIT-Hyderabad** Hyderabad, India
Graduate Researcher; advisor: Prof. Madhava Krishna
Jul 2023 – present
 - 3D representation learning:** My research focuses on learning geometric structure from incomplete and noisy point clouds for reconstruction and autonomous exploration.
 - Autonomous exploration:** For SCAGE, I developed geometric-anomaly prediction and viewpoint-selection methods to direct exploration toward poorly reconstructed regions, and evaluated the approach in simulation and on a physical robot.
 - Dynamic scene reconstruction:** I contributed to implementation and integration across UGSDF, combining SDFs, Gaussian scene representations, and image-derived depth and tracking priors.
- Center of Excellence in Visual Intelligence (CEVI), KLE Tech** Hubballi, India
Student Researcher; Research Intern (Jan – May 2023)
Aug 2021 – May 2023
 - Heritage digitization:** Under Prof. Uma Mudenagudi, I worked on a **DST-funded project** supporting conservation and digital-twin generation (IHDS: Indian Heritage in Digital Space). My research upsampled and refined the sparse, noisy point clouds produced by photogrammetric reconstruction of captured images.
 - Point-cloud upsampling:** I developed ASUR3D using local occupancy fields and co-developed TP-NoDe using topology-aware perturbations and a pretrained denoiser, producing **two first-author ICCV 2023 Workshop papers**, including one with equal contribution.

PROJECTS

- **Multi-Camera–LiDAR Calibration for an Autonomous Research Vehicle** ↗ : Developed intrinsic and extrinsic calibration for a rig comprising **six cameras and a 64-channel LiDAR**, achieving **camera-intrinsic reprojection error below 0.05 px**. Built a hybrid ChArUco–retroreflective target and implemented robust optimization across multiple frames, automatic rejection of unreliable observations, and residual diagnostics. Checked LiDAR-to-image alignment qualitatively during a campus drive.
Tech Stack: Python, OpenCV, NumPy, SciPy, Rerun
- **Dynamic-Obstacle Detection and Avoidance** ↗ : Co-developed and deployed a ROS navigation stack on a Husky A200, integrating RGB-D/LiDAR obstacle tracking, Fast-LIO localization against a prebuilt map, global planning, and MPC-based reactive avoidance. The system avoided collisions in **95% of dynamic-obstacle encounters across more than 20 test trials**.
Tech Stack: ROS, Fast-LIO, RGB-D/LiDAR perception, RRT*, MPC, RViz
- **ARROU: Adversarially Robust Rank-One Unlearning for LLMs** ↗ : Co-developed an attacker-in-the-loop framework combining latent adversarial training with localized rank-one model editing for targeted unlearning in large language models. Evaluated forgetting and retained utility on Llama 3.2 1B using WMDP and TOFU benchmarks.
Tech Stack: Python, PyTorch, Hugging Face Transformers, CUDA
- **3D Point-Cloud Instance Segmentation** ↗ : Designed a U-Net-based architecture with a custom encoder for learning global point-cloud features. On S3DIS, achieved **67.6% semantic mIoU and 53.8% instance mAP@50 on Area 5**, and **57.7% instance mAP@50 under six-fold cross-validation**.
Tech Stack: Python, PyTorch, PyTorch3D, Open3D
- **Camera-Based Road-Hazard Detection (Bosch PRIXEL)** ↗ : Co-developed a camera-based system for detecting potholes and marked or unmarked speed breakers. Collected and annotated a dataset of Indian road scenes and used it to develop and evaluate the detection pipeline.
Tech Stack: Python, PyTorch, OpenCV, ONNX
- **Learning-Based Point-Cloud Hole Filling** ↗ : Developed a model to detect and fill gaps in point clouds caused by occlusion and reflections. Designed a synthetic-hole generation pipeline to create training and evaluation examples.
Tech Stack: Python, PyTorch, PyTorch3D, Open3D
- **MAST3R-SLAM** ↗ : Extended an existing MAST3R implementation with camera-pose and point-cloud extraction utilities; inspected reconstructed geometry on KITTI sequences.
Tech Stack: Python, PyTorch

TECHNICAL SKILLS

- **Programming:** Python, C/C++, CUDA, Embedded C
- **Deep Learning:** PyTorch, ONNX, Hugging Face Transformers; LLM unlearning, model editing, adversarial training
- **3D Vision:** Representation learning, point-cloud processing, neural implicit representations, SDFs, 3D Gaussian Splatting, reconstruction, semantic and instance segmentation
- **Robotics & Sensors:** ROS, camera intrinsic/extrinsic calibration, camera–LiDAR calibration, RGB-D/LiDAR fusion, SLAM/localization, motion planning, MPC
- **Data & Tools:** Dataset collection and annotation, synthetic training-data generation, OpenCV, Open3D, PyTorch3D, NumPy, SciPy
- **Simulation & Diagnostics:** Gazebo, CARLA, Habitat, RViz, Rerun

TEACHING AND SERVICE

- **Teaching Assistant:** Mobile Robotics (CS7.503) ↗ , IIIT-Hyderabad — Monsoon 2026
- **Instructor:** RRC Summer School, IIIT-Hyderabad (2025 & 2026) — Taught deep learning for 3D vision, neural representations, and 3D Gaussian Splatting; 2026 lecture recording ↗
- **Reviewer:** ICVGIP 2026 ↗ ; SeMaNa Workshop (IROS 2026) ↗

HONORS AND AWARDS

- Certificate of appreciation in Bosch’s PRIXEL Program, Bosch Global Software Technologies — Awarded for developing a robust, real-time computer vision-based speed breaker and pothole detection model.